[INCH-POUND]
A-A-52547
February 26, 1996
SUPERSEDING
MIL-H-13444E
8 January 1986

COMMERCIAL ITEM DESCRIPTION

HOSE AND HOSE ASSEMBLIES, RUBBER: FUEL AND OIL

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers two types of fuel and oil hose and hose assemblies used as lines carrying fuel, oil or diesel fuel in vehicles and other applications.

2. CLASSIFICATION. Hose and hose assemblies are classified as follows:

Type I - Hose in accordance with Type 30R2 of SAE J30 and rated for temperature ranges from minus 65 degrees Fahrenheit (-65°F) to plus 250°F [-54 degrees Celsius (-54°C) to +121°C].

Type II - Has been discontinued (see 7.5).

Type III - Hose in accordance with Type 30R3 of SAE J30 and rated for temperature ranges from -55°F to +300°F (-48°C to +149°C).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 4720

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3. SALIENT CHARACTERISTICS

3.1 Materials. The material shall be as specified herein and in accordance with the manufacturer's material's specifications for rubber hose and hose assemblies. The use of recovered material made in compliance with regulatory requirements is acceptable provided that all other requirements of this CID are met (see 4.1).

3.1.1 Hose assembly fittings. Unless otherwise specified, the fittings shall be made from ferrous material and shall be zinc plated as per ASTM B633 Type II, Fe/zn 5 or ASTM B695, Type II, Class 5 (see 7.1).

3.2 Construction. Hose shall be constructed of seamless, smooth bore tube of flexible material, a reinforcement, and an outer cover. Hose and hose assemblies shall not have been made more than one year prior to being offered for sale to the Government.

3.2.1 Inner tube. The inner tube shall be fabricated from synthetic, oil resistant rubber polymer.

3.2.2 Reinforcement. Reinforcement of Type I hose shall be in accordance with Type 30R2 of SAE J30. Reinforcement of Type III hose shall be in accordance with Type 30R3 of SAE J30 and of such combined strength as to meet the salient characteristics of this CID.

3.2.3 Cover. The Type I hose outer cover shall be compounded from polychloroprene. The Type III hose outer cover shall consist of oil resistant, rubber impregnated fabric.

3.2.4 Fittings. Unless otherwise specified in the contract order (see 7.1), one fitting in each hose assembly shall be of the male type and one of the female type. The female type shall incorporate a swivel.

3.2.4.1 Screw threads. Screw threads of fittings shall be in accordance with ASME B1.1 for the size specified. Classes of fits for threads shall be adequate to meet all requirements of this CID without leakage or failure.

3.2.4.2 Wrench flats. Nominal distance across the wrench flats shall be in multiples of 0.063 inch (in.) [1.6 millimeter (mm)]. Tolerances shall not exceed those specified in ANSI B18.2.1.

3.2.4.3 Swivel fittings. Swivel fittings shall swivel freely with hand torque.

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3.3 Dimensions.

3.3.1 Inner tube thickness. The average thickness of tube shall not be less than 0.040 in. (1.02 mm).

3.3.2 Bulk length. Unless otherwise specified (see 7.1), bulk hose shall be furnished in random lengths up to 500 feet (ft) [152 meter (m)]. At least 80 percent of the lengths furnished shall be 40 ft (12 m) or more in length, minimum length shall be 3 ft (0.914 m) (see 7.6).

3.3.3 Cut length. When specified (see 7.1), hose shall be furnished in cut lengths with a tolerance of plus or minus one percent for lengths of 3 ft [91.4 centimeter (cm)] or greater and ±0.25 in.
(± 0.64 cm) for lengths of less than 3 ft (91.4 cm) (see 7.6).

3.3.4 Hose assembly lengths. When specified (see 7.1), lengths of hose assemblies shall be as specified in the contract or order. See A-A-52547/1 and A-A-52547/2 for more information.

3.3.5 Diameters. The inside and outside diameters shall conform to Tables I and II, for the type and size specified.

3.3.6 Concentricity. The inside diameter and the outside diameter shall be concentric with 0.030 in. (0.8 mm) full indicator movement (FIM) on 0.125 in. (3.2 mm) and 0.187 in. (4.8 mm) hose and within 0.04 in. (1.0 mm) FIM on sizes greater than 0.187 in. (4.8 mm).

3.4 Performance. The following requirements shall be applicable to hose and hose assemblies of Type I and Type III unless otherwise specified herein. The term sample shall hereafter refer to either hoses or hose assemblies of both types.

TABLE I. Inside and outside diameters (Type I).

Nominal size, I.D. in. (mm)	Inside diameter in. (mm)		Outside diameter in. (mm)	
	min.	max.	min.	max.
0.125 (3)	0.109 (2.77)	0.141 (3.58)	0.328 (8.33)	0.375 (9.53)
0.187 (5)	0.172 (4.37)	0.203 (5.16)	0.406 (10.31)	0.468 (11.89)
0.250 (6.5)	0.234 (5.94)	0.266 (6.76)	0.469 (11.91)	0.531 (13.49)
0.312 (8)	0.296 (7.52)	0.328 (8.33)	0.531 (13.49)	0.593 (15.06)
0.375 (9.5)	0.359 (9.12)	0.391 (9.93)	0.594 (15.09)	0.656 (16.66)
0.500 (12.5)	0.469 (11.91)	0.531 (13.49)	0.750 (19.05)	0.812 (20.63)
0.625 (16)	0.593 (15.06)	0.656 (16.66)	0.937 (23.80)	1.000 (25.40)
0.750 (19)	0.718 (18.24)	0.781 (19.84)	1.062 (26.98)	1.125 (28.58)

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TABLE II. Inside and outside diameters (Type III).

Nominal size, I.D. in. (mm)	Inside diameter in. (mm)		Outside diameter in. (mm)	
	min.	max.	min.	max.
0.187 (5)	0.188 (4.78)	0.216 (5.85)	0.472 (11.99)	0.510 (12.95)
0.250 (6.5)	0.250 (6.35)	0.281 (7.14)	0.535 (13.59)	0.573 (14.55)
0.312 (8)	0.313 (7.95)	0.344 (8.74)	0.597 (15.16)	0.635 (16.13)
0.406 (10.5)	0.406 (10.31)	0.437 (11.10)	0.714 (18.14)	0.760 (19.30)
0.500 (12.5)	0.500 (12.70)	0.539 (13.69)	0.808 (20.52)	0.854 (21.69)
0.625 (16)	0.625 (15.88)	0.667 (16.94)	0.933 (23.70)	0.979 (24.87)
0.875 (22)	0.875 (22.23)	0.917 (23.29)	1.175 (29.85)	1.237 (31.43)
1.125 (29)	1.125 (28.58)	1.172 (29.77)	1.456 (36.98)	1.518 (38.56)
1.375 (35)	1.375 (34.93)	1.422 (36.12)	1.691 (42.95)	1.753 (44.53)
1.813 (46)	1.812 (46.03)	1.859 (47.22)	2.096 (53.24)	2.190 (55.63)

3.4.1 Length change -Type I. When pressurized to working pressures specified in Table III, the test sample shall change no more than plus or minus 5% from its original free length. For hose assemblies, the free length shall be the length between the fittings.

3.4.2 Length change - Type III. When pressurized to working pressures specified in Table III, the test sample length increase shall be no greater than 6 percent for all sizes.

TABLE III. Pressure requirements.

Nominal hose size (I.D.) in. (mm)	Burst pressure		Proof pressure		Working pressure	
	Type I psi (kPa)*	Type III psi (kPa)	Type I psi (kPa)	Type III psi (kPa)	Type I psi (kPa)	Type III psi (kPa)
0.125 (3)	2000 (13790)	1/	1000 (6895)	1/	500 (3447)	1/
0.187 (5)	1700 (11721)	6000 (41370)	850 (5861)	3000 (20685)	425 (2930)	1500 (10342)
0.250 (6.5)	1250 (8619)	6000 (41370)	625 (4309)	3000 (20685)	300 (2068)	1500 (10342)
0.312 (8)	1100 (7584)	4000 (27580)	550 (3792)	2000 (13790)	275 (1895)	1000 (6895)
0.375 (9.5)	1000 (6895)	1/	500 (3447)	1/	250 (1724)	1/
0.406 (10.5)	1/	4000 (27580)	1/	2000 (13790)	1/	1000 (6895)
0.500 (12.5)	750 (5171)	3500 (24132)	375 (2587)	1750 (12066)	200 (1379)	875 (6033)
0.625 (16)	700 (4826)	3000 (20685)	350 (2413)	1500 (10342)	175 (1206)	750 (5171)
0.750 (19)	500 (3447)	1/	250 (1724)	1/	125 (862)	1/
0.875 (22)	1/	1500 (10342)	1/	750 (5171)	1/	375 (2587)
1.125 (29)	1/	1250 (8619)	1/	625 (4309)	1/	300 (2068)
1.375 (35)	1/	1000 (6895)	1/	500 (3447)	1/	250 (1724)
1.812 (46)	1/	750 (5171)	1/	375 (2587)	1/	200 (1379)

1/ Hose type not available in this size.
* psi = pounds per square inch & kPa = kilopascals.

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3.4.3 Proof pressure. The test samples shall withstand the proof pressure up to and including that specified in Table III without any kind of damage when a fluid (water, engine oil, or hydraulic oil) shall be forced into them as specified in ASTM D380.

3.4.4 Burst pressure. The test samples shall not burst at any pressure up to that specified in Table III when tested in accordance with ASTM D380. There shall be no damage around the fittings of test samples below the specified minimum burst pressure.

3.4.5 Low temperature flexibility. The test samples shall show no sign of damage when subjected to the low temperature flexibility test as specified in ASTM D1055 section 27-30.

3.4.6 Ozone resistance. The covers of Type I hose shall show no evidence of cracks or breaks when being subjected to the following test: a sample of 30 in. (76 cm) length shall be wrapped around a suitable mandrel (see 7.4) and secured at both ends. After conditioning in the air for 45 minutes, the sample shall be placed in a chamber containing air mixed with an ozone concentration of 50 $\pm$ 5 millipascal (mPa). The temperature of the chamber shall be 100 $\pm$ 5°F (38 $\pm$ 2.5°C). After 168 hours exposure, the cover of the sample shall be examined under 7-power magnification.

3.4.7 Vacuum collapse resistance. The decrease in the outside diameter of Type I hose shall be no greater than 15 percent when being subjected to an internal vacuum equivalent to 10 pounds per square inch (psi) [68 kilopascals (kPa)] or 20 in. (50.8 cm) of mercury for hose less than 0.5 in. (12.7 mm) nominal size. For hoses of 0.5 in. (12.7 mm) and larger, the internal vacuum shall be equivalent to 5 psi (34 kPa) or 10 in. (25.4 cm) of mercury . The vacuum shall be applied while samples bent around the suitable mandrel (see 7.4).

3.4.8 Oil resistance. When subjected to the oil resistance test specified in ASTM D1056 section 24-29; the decrease and increase in the volume of Type I hose shall be no more than 5 and 25 percent respectively. The increase in volume of the cover of Type I shall be no more than 100 percent. For Type III, the increase in volume shall be no more than 30 percent.

3.4.9 Fuel resistance. After 48 hours immersion in fuel conforming to ASTM D471 reference fuel D at a temperature of 80 $\pm$ 10°F (27 $\pm$ 5.5°C), the volume increase in Type I shall not exceed 60 and 110 percent for the hose and the cover respectively.

3.4.10 Vibration resistance. The test samples shall not leak nor shall show any evidence of damage when subjected to a vibration through an amplitude of 0.35 in. (8.7 mm) at a rate of 1800 $\pm$ 10 cycles per minute for 200 hours. During the vibration period, test fluid SAE Grade 10 W oil shall be circulated through the sample at a pressure of 45 psi (310 kPa) and at a temperature of 250 $\pm$ 5°F (121 $\pm$ 3°C).

3.4.11 Fungus resistance. The hose and hose assemblies shall be constructed of materials that will not support fungus growth.

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3.4.12 Hot oil circulation. The samples of Type III shall exhibit no leaks, ruptures, or cracks, internally or externally when subjected to SAE Grade 30 oil circulation at a temperature of 300 ± 5°F (149 ± 3°C), and at a pressure of 175 ± 25 psi (1200 ± 175 kPa) for 240 ± 1 hours in an ambient temperature of 75 ± 10°F (24 ± 5.5°C). After the circulation test, the sample shall be bent around the suitable mandrel (see 7.4) with hose at a temperature of 75 ± 10°F (24 ± 5.5°C). While bent around the mandrel, the sample shall be proof-tested in accordance with 2.4.3 and inspected.

3.4.13 Bend radius. The hose shall show no sign of collapse or any other damage when being subjected to an inside bend radius as specified in Table IV.

TABLE IV. Bend radius (hose assembly installation).

Nominal size, I.D. inch (mm)	Type I inch (mm)	Type III inch (mm)
0.125 (3)	1.75 (44.5)	1/
0.187 (5)	2.00 (50.8)	0.75 (19.1)
0.250 (6.5)	2.25 (57.2)	1.00 (25.4)
0.312 (8)	2.75 (69.9)	1.25 (31.8)
0.375 (9.5)	3.00 (76.2)	1/
0.406 (10.5)	1/	1.75 (44.5)
0.500 (12.5)	3.75 (95.3)	2.25 (57.2)
0.625 (16)	4.75 (120.7)	2.75 (69.9)
0.750 (19)	5.50 (139.7)	1/
0.875 (22)	1/	3.50 (88.9)
1.125 (29)	1/	4.50 (114.3)
1.375 (35)	1/	7.50 (190.5)
1.812 (46)	1/	14.00 (355.6)

1/ Hose type not available in this size.

3.5 Identification marking. Identification marking shall be permanent and shall include, as a minimum, the manufacturer's CAGE code and part number, and the part identification number (PIN) in accordance with CID specification sheet. In addition the Type I hose shall be marked with the capital letters “OZ” to show ozone resistance. The markings shall be repeated at intervals at intervals of no more than 12 in. (30.5 cm) along the hose. The outer cover of bulk hose or hose in hose assemblies shall be marked with a water resistant, gasoline resistant, and oil resistant, continuous yellow stripe.

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4. REGULATORY REQUIREMENTS

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.1).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. PIN and quantity required.
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
d. Type and nominal size of hose required.
e. If length of bulk hose shall be other than as specified.
f. If hose shall be furnished in cut, rather than random lengths and lengths required.
g. If hose assemblies required, and length of hose assemblies.
h. Fittings to be used on hose assemblies, if other than specified.
i. Material used in fitting, if other than specified.
j. Applicable level and packaging requirements.

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7.2 Addresses for obtaining copies of referenced documents.

7.2.1 Non-Government publications. Copies of ASME B1.1 "Unified Inch Screw Threads (UN and UNR Thread Form) (DoD Adopted)" are available from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017. Copies of ANSI B18.2.1 "Square and Hex Bolts and Screws (DoD Adopted)" are available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036. Copies of ASTM B633 "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel (DoD Adopted)", ASTM B695 "Coatings of Zinc Mechanically Deposited on Iron and Steel (DoD Adopted)", ASTM D380 "Standard Test Methods for Rubber Hose (DoD Adopted)", ASTM D1055 "Standard Specifications for Flexible Cellular Materials Latex Foam (DoD Adopted)", ASTM D1056 "Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber (DoD Adopted)" and ASTM D471 "Standard Test Method for Rubber Property - Effect of Liquids (DoD Adopted)" are available from The American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Copies of SAE J30 "Fuel and Oil Hoses (DoD Adopted)" are available from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

7.3 Cross-reference data. The hose and hose assemblies conforming to this CID are interchangeable/substitutable with the hose and hose assemblies conforming to MIL-H-13444E, MS52103D and MS52104B.

7.4 Mandrel sizes. The mandrel sizes for ozone resistance, vacuum collapse resistance and for hot oil circulation tests should be in accordance with table V.

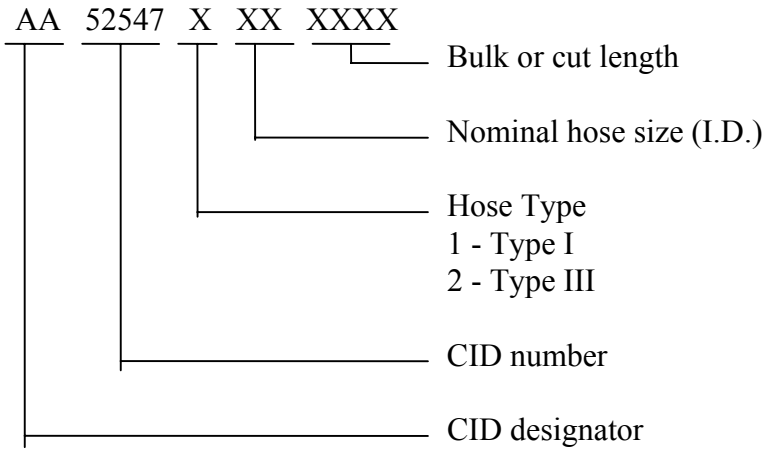
TABLE V. Reference mandrel sizes.

Hose size Inches (mm)	Paragraph 3.4.12	Paragraph 3.4.6	Paragraph 3.4.7
0.125 (3)	3.5 (89)	2.45 (62)	NA
0.187 (5)	4 (102)	2.80 (71)	2 (51)
0.250 (6.5)	4.5 (114)	3.15 (80)	2.25 (57)
0.312 (8)	5.5 (140)	3.85 (98)	2.75 (70)
0.375 (9.5)	6 (152.5)	4.20 (107)	NA
0.406 (10.5)	7 (178)	NA	4.62 (117)
0.500 (12.5)	7.5 (190.5)	5.25 (133.5)	5.50 (140)
0.625 (16)	9.5 (241)	6.65 (169)	6.50 (165)
0.750 (19)	11 (279.5)	7.70 (195.5)	NA
0.875 (22)	12 (305)	NA	7.38 (187)
1.125 (29)	15 (381)	NA	9 (229)
1.375 (35)	16 (406)	NA	11 (279)
1.812 (46)	20 (508)	NA	17 (432)

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7.5 Type II. Type II hose with "Circular Woven Fabric Reinforcement or Single Braid Reinforcement" has been discontinued from the Revision E of MIL-H-13444.

7.6 Part or identification number (PIN). The PINs to be used for bulk or cut length hose acquired to this CID are created as follows:



MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1

Review Activities:
Army - CD, ME, MI
Navy - MC
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 4720-0066)

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